1. All Dogs bark.
2. Some students study.
3. All birds can fly.
4. There is a person who loves chocolate.
5. Every student has a favorite subject.
6. Not all cats are black.
7. There exists a car that is not red.
8. No one likes to fail.
9. For every problem, there is a solution.
10. Some students love both math and art.
11. Every person who owns a cat likes to read.
12. Not all books are written by famous authors.
13. Some employees are skilled in both programming and management.
14. No bird can swim.
15. There is a professor who teaches both mathematics and computer science.

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Solution:

All Dogs bark.

$$\forall x\,(Dog(x) \rightarrow Barks(x))$$

Some students study.

$$\exists x\,(Student(x) \wedge Studies(x))$$

All birds can fly.

$$\forall x\,(Bird(x) \rightarrow CanFly(x))$$

There is a person who loves chocolate.

$$\exists x\,(Person(x) \wedge Loves(x, Chocolate))$$

Every student has a favorite subject.

$$\forall x\,(Student(x) \rightarrow \exists y\,(Subject(y) \wedge HasFavoriteSubject(x, y)))$$

Not all cats are black.

$$\neg(\forall x\,(Cat(x) \rightarrow Black(x)))$$

There exists a car that is not red.

$$\exists x\,(Car(x) \wedge \neg Red(x))$$

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No one likes to fail.

$$\neg\,\exists x\,(Person(x) \wedge Likes(x, Fail))$$

For every problem, there is a solution.

$$\forall x\,(Problem(x) \rightarrow \exists y\,(Solution(y, x)))$$

Some students love both math and art.

$$\exists x\,(Student(x) \wedge Loves(x, Math) \wedge Loves(x, Art))$$

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Every person who owns a cat likes to read.

$$\forall x\,(\text{Person}(x) \land \exists y\,(\text{Cat}(y) \land \text{Owns}(x, y)) \rightarrow \text{Likes}(x, \text{Read}))$$

Not all books are written by famous authors.

$$\neg(\forall x\,(\text{Book}(x) \rightarrow \exists y\,(\text{FamousAuthor}(y) \land \text{WrittenBy}(x, y))))$$

Some employees are skilled in both programming and management.

$$\exists x\,(\text{Employee}(x) \land \text{SkilledIn}(x, \text{Programming}) \land \text{SkilledIn}(x, \text{Management}))$$

No bird can swim.

$$\forall x\,(\text{Bird}(x) \rightarrow \neg\text{Swims}(x))$$

There is a professor who teaches both mathematics and computer science.

$$\exists x\,(\text{Professor}(x) \land \text{Teaches}(x, \text{Mathematics}) \land \text{Teaches}(x, \text{ComputerScience}))$$

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Define Bayes rule for probabilistic problem. If probability of symptoms on patients is 1/20 and probability of disease known by doctor is 1/45000. Also the probability of symptoms condition to disease is 50%. Now, calculate the probability of disease condition to symptoms using Bayes theorem.

Bayes Rule: It's a mathematical formula for calculating conditional probabilities.
It allows us to update our beliefs about the likelihood of an event (hypothesis) based on new evidence.

The formula is:

P(A | B) = (P(B | A) * P(A)) / P(B)

Where:
P(A | B) is the probability of event A occurring given that event B has occurred (posterior probability)
P(B | A) is the probability of event B occurring given that event A has occurred (likelihood)
P(A) is the prior probability of event A occurring
P(B) is the prior probability of event B occurring

Applying Bayes Rule to the Problem:

Event A: Having the disease
Event B: Having the symptoms

Given Information:
P(B) = 1/20 (probability of symptoms)
P(A) = 1/45000 (probability of disease)
P(B | A) = 0.5 (probability of symptoms given the disease)

Applying Bayes Rule:

P(A | B) = (P(B | A) * P(A)) / P(B)

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In a survey, One percent of women over 50 have breast cancer. Ninety percent of women who have breast cancer test positive on mammograms. Eight percent of women will have false positives. From the above statistics, apply Bayes theorem to find the probability that a woman has cancer if she has a positive mammogram result.

Define Events:

- A: Having breast cancer
- B: Receiving a positive mammogram result

Gather Probabilities:

- P(A): Prior probability of having breast cancer = 1% = 0.01
- P(B | A): Probability of a positive mammogram given breast cancer = 90% = 0.90
- P(B | !A): Probability of a positive mammogram given no breast cancer (false positive) = 8% = 0.08

Calculate P(B): Probability of getting a positive mammogram regardless of cancer status
This involves accounting for both true positives and false positives:

P(B) = P(B | A) * P(A) + P(B | !A) * P(!A)
where:

- P(!A): Probability of not having breast cancer = 1 - P(A) = 1 - 0.01 = 0.99

Apply Bayes' theorem:
Now we can calculate the probability of having breast cancer given a positive mammogram result (P(A | B)):
P(A | B) = (P(B | A) * P(A)) / P(B)

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A Spam detector system detects spam emails. Assume that the word "offer" occurs in 80% of the spam messages in your account and in 10% of your non-spam emails 30% of the received emails are considered a spam. Now, you will receive a new message which contains the word "offer". What is the probability that it is a spam?

Define Events:

- A: Email is spam
- B: Email contains the word "offer"

Gather Probabilities:

- P(A): Prior probability of an email being spam = 30% = 0.30
- P(B | A): Probability of an email containing "offer" given it's spam = 80% = 0.80
- P(B | !A): Probability of an email containing "offer" given it's not spam = 10% = 0.10

Calculate P(B): Probability of an email containing "offer" regardless of spam status:

$$P(B) = P(B \mid A) * P(A) + P(B \mid !A) * P(!A)$$

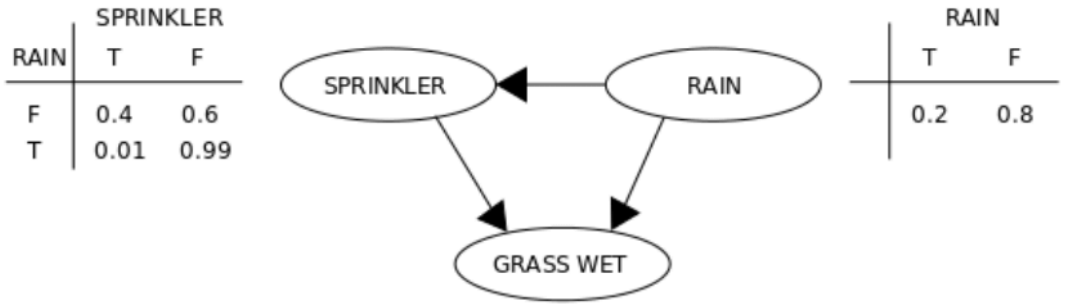
Apply Bayes' theorem:

$$P(A \mid B) = (P(B \mid A) * P(A)) / P(B)$$

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Consider the given Bayesian network with conditional probability tables and find the probability of the events grass is wet, sprinkler is on and it is raining.



RAIN	SPRINKLER T	SPRINKLER F
F	0.4	0.6
T	0.01	0.99

RAIN T	RAIN F
0.2	0.8

SPRINKLER	RAIN	GRASS WET T	GRASS WET F
F	F	0.0	1.0
F	T	0.8	0.2
T	F	0.9	0.1
T	T	0.99	0.01

Identify Relevant Probabilities:

From the conditional probability tables in the image, note down the following probabilities:

- P(Rain) = 0.2
- P(Sprinkler = On | Rain = True) = 0.01
- P(GrassWet = True | Sprinkler = On, Rain = True) = 0.99

Apply the Chain Rule:

To find the joint probability of all three events (grass wet, sprinkler on, rain), use the chain rule:

P(GrassWet = True, Sprinkler = On, Rain = True) = P(Rain = True) * P(Sprinkler = On | Rain = True) * P(GrassWet = True | Sprinkler = On, Rain = True)

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Assume the following facts:

- **John likes all kinds of food.**
- **Apple and vegetable are food**
- **Anything anyone eats and is not killed is food.**
- **Anil eats peanuts and still alive**
- **Harry eats everything that Anil eats.**
- **Anyone who's not killed means, he is alive.**
- **Represent these facts in predicate logic.**

Using Resolution prove “John likes Peanuts”

Solution:

Convert to FOL:

John likes all kinds of food.

$\forall x$ (Food(x) → Likes(John, x))

Apple and vegetable are food.

Food(Apple) ∧ Food(Vegetable)

Anything anyone eats and is not killed is food.

$\forall x \forall y$ ((Eats(y, x) ∧ ¬Killed(y)) → Food(x))

Anil eats peanuts and is still alive.

Eats(Anil, Peanuts) ∧ ¬Killed(Anil)

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Harry eats everything that Anil eats.

$\forall x$ (Eats(Anil, x) → Eats(Harry, x))

Anyone who's not killed means he is alive.

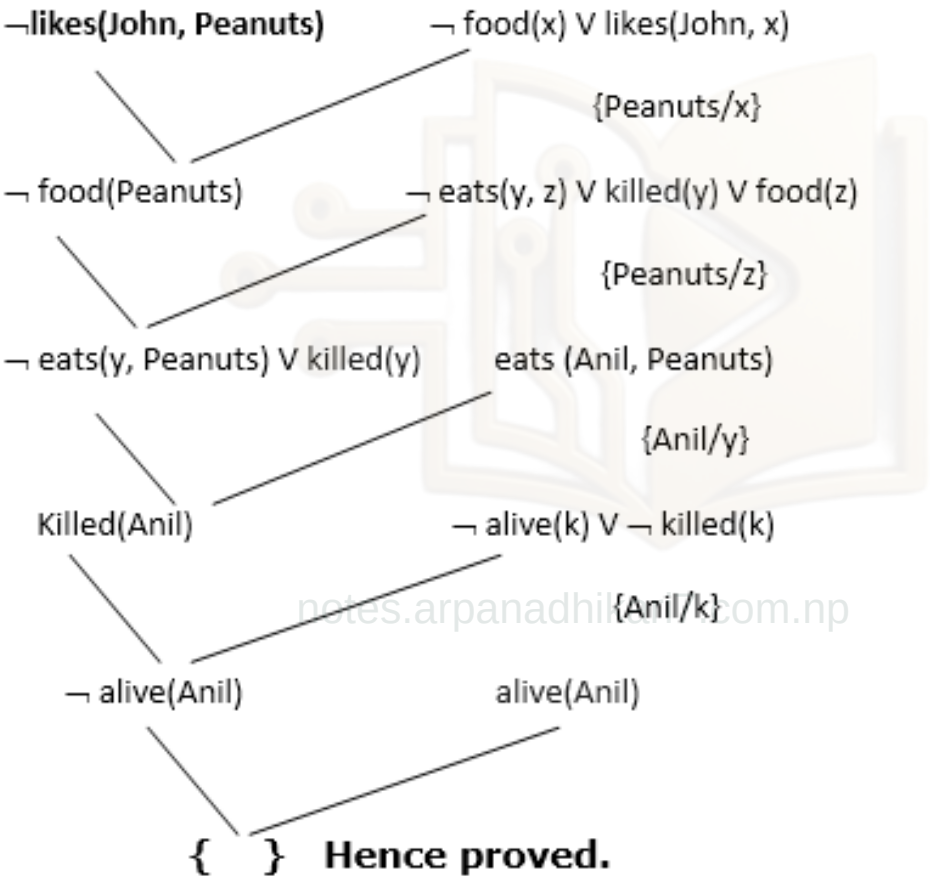
$\forall x$ (¬Killed(x) → Alive(x))
$\forall x$ (Alive(x) → ¬Killed(x))

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Convert to CNF

- ¬Food(x) ∨ Likes(John, x)
- ¬Eats(y, z) ∧ Killed(y) ∨ Food(z)
- Food(Apple) ∧ Food(Vegetable)
- Eats(Anil, Peanuts) ∧ ¬Killed(Anil)
- ¬Eats(Anil, w) ∨ Eats(Harry, w)
- Killed(g) ∨ Alive(g)
- ¬Killed(k) ∨ ¬Alive(k)

¬Likes(John, Peanuts)



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Assume the following facts:

- **Ram likes all kinds of food.**
- **Orange is food**
- **Rice is food.**
- **Anything anyone eats and is not killed by is food**
- **Krishna eats popcorn and is still alive.**
- **Radha eats anything Krishna eats.**

Prove that Ram likes popcorn using resolution.

Solution:

- $\forall x(Food(x) \rightarrow Likes(Ram,x))$
- Food(Orange)
- Food(Rice)
- $\forall x \forall y(Eats(x,y) \wedge \neg Killed(x) \rightarrow Food(y))$
- $Eats(Krishna,popcorn) \wedge Alive(Krishna)$
- $\forall x(Eats(Radha,x) \rightarrow Eats(Krishna,x))$

$\neg Likes(Ram,popcorn)$

Convert to CNF and draw resolution tree for given statement.

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